

Soul and land pollution LQs

2023

(1)

The solid components of soil consist of mineral particles and organic matter. These components form the foundation of soil structure and significantly influence its properties and fertility. Here's a detailed description of these solid components:

Mineral Particles

1. Sand:

Sand particles are large and irregular in shape, which means they don't pack closely together. This creates large spaces, or pores, between the particles.

Because of these large pores, sandy soils have excellent drainage and aeration. However, they also have low water and nutrient retention capabilities, meaning they dry out quickly and may require more frequent watering and fertilization.

2. Silt:

Silt particles are smaller and smoother than sand, which allows them to pack together more tightly than sand but not as tightly as clay.

Silty soils have moderate water-holding capacity and good nutrient retention. They are typically fertile and easy to work with but can be prone to erosion if not managed properly.

3. Clay:

Clay particles are extremely small and flat, with a large surface area relative to their size. This allows them to hold onto water and nutrients very effectively.

Clay soils can retain significant amounts of water and nutrients, making them very fertile. However, they drain poorly and can become waterlogged. Clay can also become very hard when dry, making it difficult for plant roots to penetrate.

Organic Matter

1. Humus:

Humus forms through the decomposition of plant and animal matter by soil microorganisms. It is dark, spongy, and stable, meaning it decomposes very slowly.

Humus improves soil structure by binding mineral particles into aggregates, which enhances soil porosity and water-holding capacity. It also increases the soil's ability to retain and supply nutrients to plants, promotes beneficial microbial activity, and helps buffer soil pH.

2. Plant and Animal Residues:

These are less decomposed than humus and include a variety of materials like leaves, stems, roots, and the remains of soil fauna.

As these residues decompose, they contribute to the soil's organic matter content. They provide a continuous supply of nutrients, improve soil structure, and support a diverse and active microbial community.

The solid components of soil—mineral particles (sand, silt, and clay) and organic matter—determine its texture, structure, and fertility. Soil texture, defined by the proportions of these particles, affects drainage and nutrient-holding properties, with loam being ideal for plant growth. Soil structure, the arrangement of particles into aggregates, enhances water infiltration, root penetration, and aeration. Fertility is influenced by the combination of mineral particles and organic matter, particularly humus, which is crucial for nutrient cycling and availability. These interactions are essential for soil health and plant productivity, influencing water retention, nutrient availability, and soil aeration.

(2) {Also repeated 2021}

Industries contribute to soil pollution through various activities and processes, including:

(Industries are major source of soil pollution due to following reasons:)

- **Industrial Waste Disposal:** Improper disposal of industrial waste, such as chemicals, heavy metals, and toxic substances, directly contaminates soil when not managed correctly.
- **Accidental Spills and Leaks:** Accidental spills and leaks from storage tanks, pipelines, or during transportation can release hazardous materials into the soil, leading to contamination.
- **Atmospheric Deposition:** Industries emit pollutants into the air through smokestacks and other emissions. These pollutants can settle onto soil surfaces through precipitation (acid rain) or direct deposition, contaminating the soil.
- **Improper Handling of Hazardous Materials:** Improper handling, storage, or disposal of hazardous materials within industrial facilities can lead to leaks, spills, or leaching into the soil.
- **Landfills and Waste Dumps:** Industrial activities generate large volumes of solid waste, including non-biodegradable materials and by-products. Improperly managed landfills or waste dumps can release contaminants into the soil through leachate.
- **Chemical Usage:** Industries often use chemicals in manufacturing processes, such as solvents, pesticides, and fertilizers. Improper application or spillage of these chemicals can result in soil contamination.
- **Mining and Extraction:** Mining activities can release heavy metals and other pollutants into the soil, altering its composition and potentially rendering it unsuitable for agriculture or natural habitats.
- **Urbanization and Infrastructure Development:** Expansion of industrial zones, construction activities, and urban sprawl can disturb natural soil layers, leading to soil compaction, erosion, and increased vulnerability to pollution.

- **Industrial Runoff:** Stormwater runoff from industrial sites can carry pollutants such as oils, metals, and chemicals into nearby soils and water bodies, especially if proper runoff management practices are not in place.

Overall, industrial activities have the potential to significantly impact soil quality and health through direct contamination, improper waste management, and the release of pollutants into the environment. Mitigating these impacts requires stringent regulations, responsible waste management practices, and sustainable industrial processes.

2022

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Soils contain both liquid and gaseous components that play crucial roles in supporting plant growth and various ecological processes:

Liquid Components of Soil:

1. Soil Water:

Soil water exists in several forms, each with distinct characteristics and roles:

- **Gravitational Water:** Found in larger soil pores, it drains quickly due to gravity after rainfall or irrigation. While it is readily available to plants, it doesn't remain in the soil for long.
- **Capillary Water:** Retained in the smaller soil pores by surface tension, this water is crucial for plant uptake. It is the primary source of water for plants between rainfall or irrigation events.
- **Hygroscopic Water:** This is a thin film of water tightly bound to soil particles and is generally not available to plants due to the strong adhesive forces.
- **Role:** Soil water is indispensable for plant growth as it dissolves and transports nutrients from the soil to plant roots. It facilitates various biochemical reactions necessary for plant and microbial life and helps maintain cell turgor pressure, which is critical for plant structure and function. Additionally, soil water plays a role in regulating soil temperature, which is important for both plant roots and soil organisms.

Gaseous Components of Soil:

1. Soil Air:

Soil air occupies the spaces between soil particles that are not filled with water. It consists primarily of:

- **Nitrogen (N₂):** Making up about 78% of soil air, similar to the atmosphere, and it is relatively inert in the soil.
- **Oxygen (O₂):** Constitutes around 21% of soil air, less than in the atmosphere due to consumption by roots and soil microbes.

- **Other Gases:** Trace amounts of other gases, including methane and nitrous oxide, are present, especially under anaerobic conditions.

- **Role:** Oxygen in soil air is crucial for the aerobic respiration of plant roots and soil microorganisms, which is essential for energy production, nutrient uptake, and overall plant health. Adequate aeration prevents the accumulation of toxic gases and supports the decomposition of organic matter, promoting nutrient cycling and soil fertility.

2. Soil Carbon Dioxide (CO₂):

Produced by the respiration of plant roots and soil organisms, as well as through the decomposition of organic matter, CO₂ can accumulate in soil pores, especially in poorly aerated soils.

- **Role:** CO₂ influences soil pH and acts as a carbon source for certain soil microbes. While low to moderate levels of CO₂ can stimulate root and microbial activity, excessive CO₂ can lead to soil acidification and negatively impact soil health. Proper balance and ventilation are needed to maintain optimal conditions for plant growth and microbial processes.

(2)

Mining activities contribute to soil pollution primarily through the following mechanisms:

- **Release of Heavy Metals and Toxic Substances:** Mining operations often involve extracting metals and minerals that contain toxic elements such as lead, mercury, arsenic, and cadmium. These substances can leach into the soil through runoff or direct discharge, contaminating nearby land.
- **Acid Mine Drainage:** When sulfide minerals in mined rocks and exposed ores react with water and oxygen, they can generate acidic runoff known as acid mine drainage (AMD). AMD releases dissolved metals into soil and waterways, increasing soil acidity and contaminating surrounding areas.
- **Contamination from Tailings:** Tailings are finely ground rock particles left over after minerals are extracted from ore. They often contain residual metals and chemicals used in the extraction process. Improper storage or management of tailings can lead to their dispersion into the environment, contaminating soil and water sources.
- **Soil Erosion and Sedimentation:** Mining activities disturb large areas of land, leading to soil erosion and sedimentation of nearby water bodies. Eroded soil can carry pollutants such as heavy metals and chemicals into rivers and streams, impacting aquatic ecosystems and downstream communities.
- **Loss of Soil Structure and Fertility:** The physical disturbance caused by mining, including excavation and removal of topsoil, disrupts soil structure and reduces fertility. Reclamation efforts may not fully restore soil quality, leaving behind degraded land unsuitable for agriculture or natural habitats.
- **Residual Chemicals and Wastes:** Chemicals used in mining processes, such as cyanide and sulfuric acid, can persist in soil even after mining operations cease. These chemicals pose long-term risks to soil health and may continue to leach into groundwater or surface water.
- **Indirect Impacts on Biodiversity:** Soil pollution from mining can indirectly affect biodiversity by altering habitat conditions for plants and soil organisms. Loss of biodiversity further compromises ecosystem resilience and function.

Overall, mining activities can have significant and long-lasting impacts on soil quality and environmental health, requiring stringent regulations, monitoring, and sustainable practices to mitigate pollution and ensure land restoration.

2021

(2) {Repeated in 2020}

On page no. 150 (Haq nawaz bhatti)

2020

(1)

On page no. 165

2019

(a)

From notes

(b)

Soil erosion is the process by which the top layer of soil is removed by natural forces such as wind, water, or human activities. This can lead to the loss of fertile land, reduced agricultural productivity, and environmental degradation.

Causes of Soil Erosion

- 1. Water:** Rainfall, surface runoff, and river flow can wash away soil, particularly on slopes.
- 2. Wind:** In arid and semi-arid regions, wind can blow away loose, dry soil particles.
- 3. Human Activities:** Deforestation, overgrazing, improper agricultural practices, and construction can disrupt the soil structure and increase its susceptibility to erosion.

Methods of Prevention

1. Planting Vegetation:

- Vegetation, including trees, shrubs, and grasses, helps hold the soil together with their roots.
- Plants reduce the velocity of wind and water flow, significantly reducing soil erosion. The roots bind the soil, making it more resistant to erosion.

2. Terracing:

- This involves creating stepped levels on steep terrain.
- Terracing reduces the speed of water runoff, which helps in minimizing soil erosion. It also creates flat areas that can be used for agriculture.

3. Contour Plowing:

- Plowing along the contours of the land, rather than up and down slopes.
- This method reduces water runoff and encourages water infiltration into the soil, reducing erosion.

4. Cover Crops:

- Growing crops like clover or rye in between main crop seasons.
- Cover crops protect the soil from erosion by providing cover, improving soil structure, and adding organic matter.

5. Windbreaks:

- Planting rows of trees or shrubs to act as barriers against the wind.
- Windbreaks reduce wind speed over the soil surface, thus preventing wind erosion.

6. Mulching:

- Applying a layer of organic or inorganic material on the soil surface.
- Mulching reduces soil temperature fluctuations, retains moisture, and protects the soil from direct impact of raindrops, which can cause erosion.

7. No-Till Farming:

- A farming practice where the soil is not plowed, and crops are planted directly into the residue of previous crops.
- No-till farming preserves soil structure, reduces erosion, and improves water infiltration and retention.

8. Retention Structures:

- Building structures such as check dams, silt fences, and sediment traps.
- These structures slow down water flow and capture eroded soil, preventing it from being washed away.

2018

(2)

Mining

Mining is the process of extracting valuable minerals, metals, and other geological materials from the Earth. These resources are crucial for various industrial applications, including construction, manufacturing, and energy production.

The primary purpose of mining is to obtain raw materials that are essential for human activities and economic development. These materials include metals like gold, silver, and copper, as well as non-metallic resources like coal, diamonds, and limestone.

Types of Mining:

1. Surface Mining:

- Open-Pit Mining: Excavating a large pit to extract ore found near the surface. Used for metals like copper and iron.
- Strip Mining: Removing strips of soil and rock to expose ore. Commonly used for coal and lignite.
- Mountaintop Removal: Blasting away the tops of mountains to access mineral seams. Predominantly used for coal.

2. Underground Mining:

- Room and Pillar Mining: Creating rooms of ore while leaving pillars to support the roof.
- Longwall Mining: A mechanized shearer extracts ore along a continuous face.
- Block Caving: Mining by allowing ore bodies to collapse under their weight, used for deep, low-grade ores.

Environmental Impacts:

- Habitat Destruction:

Mining activities often require clearing large areas of vegetation, leading to habitat loss for many species. This can result in reduced biodiversity and disruption of ecosystems. For example, deforestation for mining can destroy the homes of many animals and plants.

- Water Pollution:

Chemicals used in mining processes, such as cyanide and mercury, can leach into water sources, contaminating rivers, lakes, and groundwater. This pollution can harm aquatic life and make water unsafe for human consumption. Acid mine drainage, a phenomenon where sulfide minerals exposed during mining produce sulfuric acid, further contaminates water bodies.

- Air Pollution:

Dust and emissions from mining operations can release particulate matter and greenhouse gases into the atmosphere. This air pollution can cause respiratory problems in nearby communities and contribute to climate change. For instance, coal mining releases methane, a potent greenhouse gas.

- Soil Erosion:

The removal of vegetation and topsoil during mining increases the risk of soil erosion. This can lead to the degradation of land, making it unsuitable for agriculture or natural vegetation regrowth. Eroded soil can also pollute waterways and disrupt aquatic habitats.

- **Acid Mine Drainage:**

When sulfide minerals in mined rock are exposed to air and water, they produce sulfuric acid. This acid can drain into nearby water bodies, lowering their pH and harming aquatic life. Acid mine drainage can persist long after mining operations have ceased, causing long-term environmental damage.

Littering

Littering is the act of improperly disposing of waste products in an inappropriate location, such as on streets, sidewalks, public places, natural environments, or bodies of water, rather than using designated waste disposal facilities like trash cans or recycling bins.

-Types of Litter:

- *Plastics:* Bottles, bags, straws, and packaging.
- *Paper Products:* Newspapers, wrappers, and napkins.
- *Metals:* Aluminum cans, foil, and other metal scraps.
- *Glass:* Bottles, jars, and broken glass.
- *Organic Waste:* Food scraps, yard waste.
- *Cigarette Butts:* One of the most common forms of litter.

-Sources of Litter:

- **Individual Actions:** Improper disposal of small items like wrappers, cans, and cigarette butts.
- **Commercial Activities:** Waste from businesses, such as packaging and promotional materials.
- **Accidental Littering:** Waste blown from overfilled bins or vehicles.

Environmental Impacts of Littering

1. Wildlife Harm:

- Animals can ingest or become entangled in litter, leading to injury or death. For example, sea turtles often mistake plastic bags for jellyfish, their natural prey.
- Birds, fish, and mammals can suffer from digestive blockages or malnutrition if they consume litter.

2. Ecosystem Disruption:

- Litter can disrupt natural habitats, leading to the displacement of species and loss of biodiversity.
- Chemicals from decomposing litter can contaminate soil and water, affecting plant and animal life.

3. Pollution:

- *Land Pollution:* Accumulation of litter can lead to unsanitary conditions, attracting pests and leading to the spread of diseases.
- *Water Pollution:* Litter can enter waterways, leading to the contamination of rivers, lakes, and oceans. This not only affects marine life but also human health, especially where water is used for drinking or recreation.
- *Air Pollution:* Burning of litter, especially plastics, can release toxic fumes and particulate matter into the air.

4. Aesthetic and Social Impact:

- Litter reduces the visual appeal of natural landscapes and urban areas, negatively affecting tourism and property values.
- It can contribute to a sense of community neglect and lower the quality of life for residents.

5. Economic Costs:

- Governments and local authorities spend significant amounts of money on cleaning up litter.
- Littering can harm businesses, especially in tourism and hospitality, by deterring visitors.

6. Long-term Environmental Impact:

- *Microplastics:* Larger plastic items break down into microplastics, which persist in the environment and enter the food chain, posing long-term health risks to wildlife and humans.
 - *Climate Change:* Decomposition of organic waste in landfills produces methane, a potent greenhouse gas contributing to climate change.
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2017

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On page no. 178 (Haq nawaz bhatti)

(5)

Mining can be a significant source of pollution, impacting the environment in various ways:

1. **Water Pollution:** Mining activities often involve the use of chemicals such as cyanide and sulfuric acid to extract minerals from ores. These chemicals can leach into nearby water bodies, leading to contamination. Additionally, mining operations can produce acid mine drainage (AMD), where sulfuric acid and heavy metals like lead, arsenic, and mercury are released into streams and rivers. This pollution not only harms aquatic life but also affects water quality for drinking and agricultural purposes.

2. **Air Pollution:** Mining operations generate dust and particulate matter, especially during activities like drilling, blasting, and crushing. These particles can contain harmful substances such as silica, which can cause respiratory diseases when inhaled by workers and nearby communities. Diesel engines and blasting operations also emit pollutants like nitrogen dioxide (NO₂) and sulfur dioxide (SO₂), contributing to air pollution and acid rain formation.
3. **Soil and Land Degradation:** Open-pit mining and the clearing of vegetation for mining infrastructure can lead to soil erosion, loss of soil fertility, and disruption of natural ecosystems. The removal of topsoil and overburden exposes underlying rocks and minerals, altering the landscape and making it prone to erosion. This degradation can have long-term effects on local flora and fauna, as well as on the livelihoods of communities dependent on agriculture and natural resources.
4. **Ecosystem Disruption:** Mining activities often result in habitat destruction and fragmentation, displacing wildlife and altering ecological dynamics. Habitats crucial for biodiversity conservation may be lost or degraded, affecting species diversity and ecosystem resilience.
5. **Global Environmental Impact:** Beyond local effects, mining contributes to global environmental challenges such as climate change. Extraction and processing of minerals require significant energy inputs, often sourced from fossil fuels, which contribute to greenhouse gas emissions. Deforestation and land use changes associated with mining also release carbon stored in forests, further exacerbating climate impacts.

Addressing these environmental impacts requires adopting sustainable mining practices, implementing effective environmental regulations, and promoting responsible resource management. Technologies for reducing water and air pollution, reclaiming mined lands, and improving waste management are critical in mitigating mining's adverse effects on the environment.

2016

(1)

Micronutrients are essential elements required by plants in small quantities for their growth, development, and overall health. These nutrients play critical roles in various physiological processes within plants, despite being needed in relatively minor amounts compared to macronutrients like nitrogen, phosphorus, and potassium. Here's a brief overview of micronutrients in soil:

1. **Types of Micronutrients:** Micronutrients essential for plant growth include iron (Fe), manganese (Mn), zinc (Zn), copper (Cu), boron (B), molybdenum (Mo), and chlorine (Cl). Each micronutrient serves specific functions in plant metabolism, enzyme activation, and cellular processes.
2. **Plant Requirements:** While required in smaller quantities compared to macronutrients, deficiencies or imbalances in micronutrient availability can severely limit plant growth and productivity. Symptoms of micronutrient deficiencies vary among plants but often manifest as chlorosis (yellowing of leaves), stunted growth, and reduced yield.
3. **Soil Availability:** The availability of micronutrients in soil depends on several factors, including soil pH, organic matter content, cation exchange capacity (CEC), and soil texture. Soil pH influences

micronutrient solubility; for instance, iron and manganese availability decreases in alkaline soils, while zinc availability decreases in highly alkaline or highly acidic soils.

4. **Role of Cation Exchange:** Micronutrients are typically cations and are subject to the same processes of cation exchange as other nutrients. Soil particles, such as clay minerals and organic matter, hold onto cations and release them into the soil solution for plant uptake as needed.
5. **Management and Applications:** To address micronutrient deficiencies, farmers may apply micronutrient fertilizers or amend soils with organic matter to improve micronutrient availability. However, excessive application can lead to toxicity issues, underscoring the importance of soil testing and targeted nutrient management practices.

In conclusion, while micronutrients are required in smaller quantities, their availability and balance in soil are crucial for optimal plant growth and productivity. Understanding soil-plant interactions and employing appropriate management practices are essential for maintaining adequate micronutrient levels and ensuring sustainable agricultural production.

(3) {Repeated in 2015}

On page no. 165 (Haq nawaz bhatti)

(5) {Repeated in 2015, 14}

On page no. 181 (Haq nawaz bhatti)

2014

(3)

Plants assimilate nutrients from the soil through their roots via several key mechanisms:

Absorption of Water and Nutrients:

- *Osmosis:* This is the movement of water across a semi-permeable membrane from a region of low solute concentration to a region of high solute concentration. Plant roots absorb water from the soil through osmosis, which also helps in the uptake of dissolved nutrients.

- *Active Transport:* This process requires energy (ATP) to move nutrients against their concentration gradient into the root cells. Transport proteins in the root cell membranes facilitate this uptake of essential nutrients like nitrogen, phosphorus, and potassium.

Diffusion:

- Nutrients in the soil solution move from areas of higher concentration (soil) to areas of lower concentration (near the root surface) until equilibrium is reached. This passive movement helps in the continuous supply of nutrients to the roots.

Mass Flow:

- As plants transpire, water is drawn from the soil into the roots and up through the plant. This movement of water also carries dissolved nutrients (such as nitrates and sulfates) towards the root surface, enhancing nutrient uptake.

Root Interception:

- As roots grow and extend through the soil, they come into direct contact with soil particles and nutrients. This physical interception allows roots to access nutrients that may not be readily available in the soil solution.

Role of pH in Nutrient Assimilation

Nutrient Solubility:

- *Acidic Soils (pH < 6):* In highly acidic soils, the solubility of micronutrients like iron (Fe), manganese (Mn), zinc (Zn), and copper (Cu) increases, making them more available to plants. However, excessive availability can lead to toxicity.
- *Neutral to Slightly Acidic Soils (pH 6-7):* This pH range is optimal for the availability of most macronutrients (nitrogen, phosphorus, potassium) and some micronutrients. Plants generally grow best in this pH range.
- *Alkaline Soils (pH > 7):* In alkaline soils, the availability of micronutrients like iron, manganese, and zinc decreases, potentially leading to deficiencies.

Toxicity:

- Low pH (highly acidic conditions) can increase the solubility of aluminum (Al) and manganese (Mn) to toxic levels, which can harm root growth and function.

Microbial Activity:

- Soil pH affects the population and activity of microorganisms, which are essential for processes like nitrogen fixation, organic matter decomposition, and nutrient cycling. Optimal microbial activity usually occurs in neutral to slightly acidic soils.

Cation Exchange Capacity (CEC) in Nutrient Assimilation

Nutrient Holding Capacity:

- Soils with high CEC have more negatively charged sites that can hold positively charged ions (cations) like potassium (K^+), calcium (Ca^{2+}), and magnesium (Mg^{2+}). This capacity helps in retaining essential nutrients within the root zone and makes them available for plant uptake over time.

Buffering Capacity:

- High CEC soils can buffer against sudden changes in soil pH by adsorbing and releasing cations. This buffering capacity helps maintain a stable pH environment, which is crucial for optimal nutrient availability and plant health.

Soil Fertility:

- Soils with low CEC, such as sandy soils, have limited ability to retain nutrients, leading to faster nutrient leaching and requiring more frequent fertilization. In contrast, soils with high CEC, such as clay and organic-rich soils, retain nutrients better and provide a more consistent supply to plants.

In summary, understanding the mechanisms of nutrient absorption, the role of pH, and the significance of CEC helps in managing soil fertility and optimizing plant growth. Proper soil management practices, such as adjusting pH and improving soil organic matter, can enhance nutrient availability and assimilation by plants.
